Task 1 Report: Differential Drive Robot Control

1. Control Strategy

Pure Pursuit

- Steers using heading error to the active waypoint.

P Controller (Proportional)

- Uses proportional distance for linear speed and proportional heading error for angular speed.

Stanley

- Combines heading alignment and cross-track correction relative to the current path segment.
- Better path-centering behaviour, especially when the robot drifts off the segment.

MPC (Sampling-Based)

- Samples control rollouts over a short horizon, predicts future states, and selects the first control from the best rollout.
- More flexible but more sensitive to horizon length, weights, and sampling bounds.

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2. Implementation Details

Parameters Used

Parameter	Value	Unit	Purpose
Linear Velocity	0.3	m/s	Forward speed
Lookahead Distance	0.5	m	Path tracking sensitivity
Waypoint Tolerance	0.1	m	Distance threshold to reach waypoint

Waypoint Sequence

Waypoint	Position
T1	[3, 0]
T2	[6, 4]
T3	[3, 4]
T4	[3, 1]
T5	[0, 3]

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3. Challenges Encountered

Challenge 1: Trade-off Between Fast Convergence and Smooth Motion

All controllers needed tuning to avoid the same issue: either approaching waypoints too slowly or overshooting them.

What was done:

- Reduced speed near waypoints.
- Limited angular commands.
- Used conservative default gains/weights.

Challenge 2: MPC Looping/Revisiting Nearby Waypoints

Compared to the other controllers, MPC initially showed repeated passes near the same waypoint region.

Main reason:

- Prediction horizon and cost weighting were not balanced enough, so local rollouts could prefer short-term improvements that looked unstable globally.

What was done:

- Increased planning horizon quality.
- Rebalanced position/heading/control penalties.
- Added smoother behaviour near waypoint transitions.

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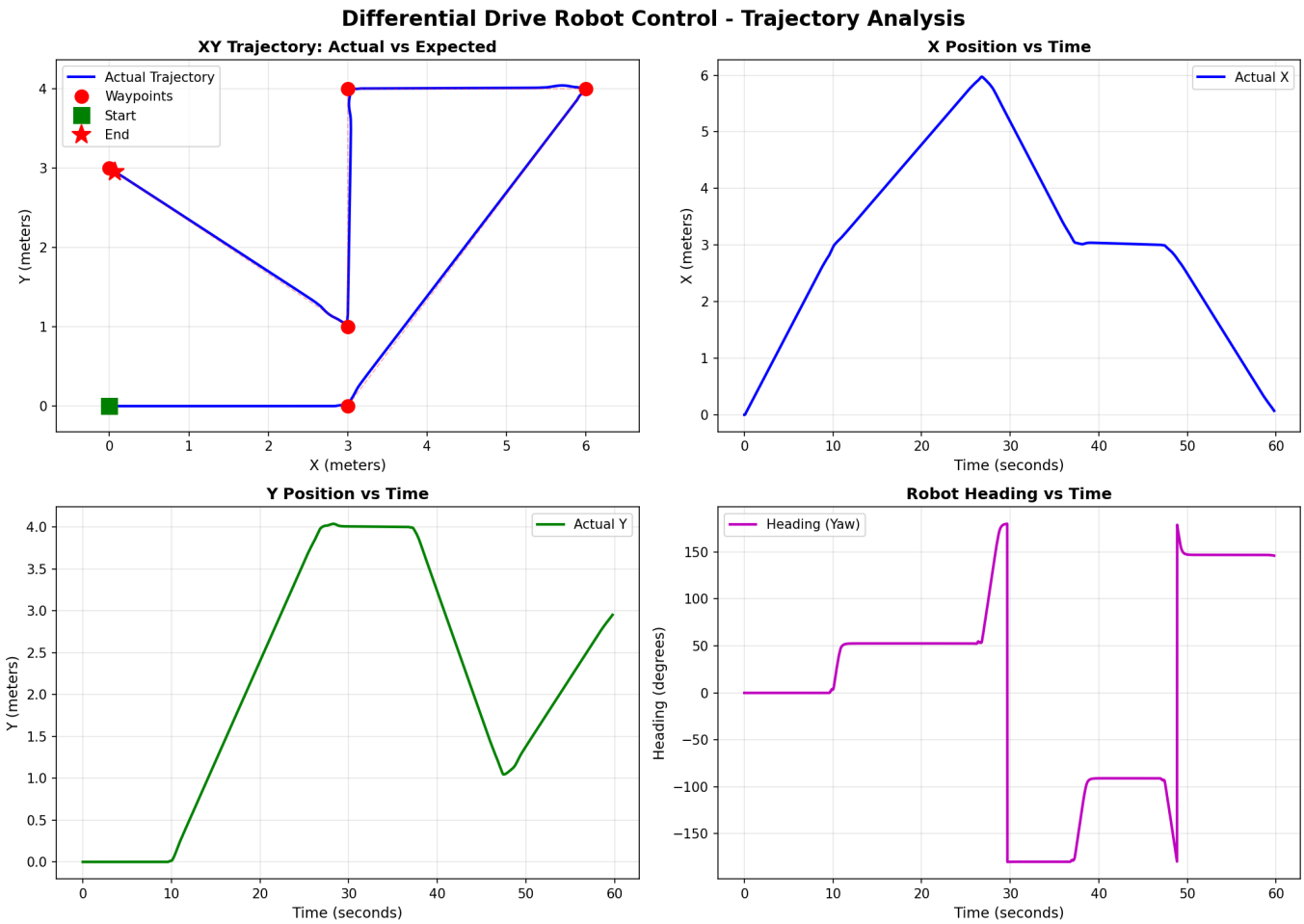
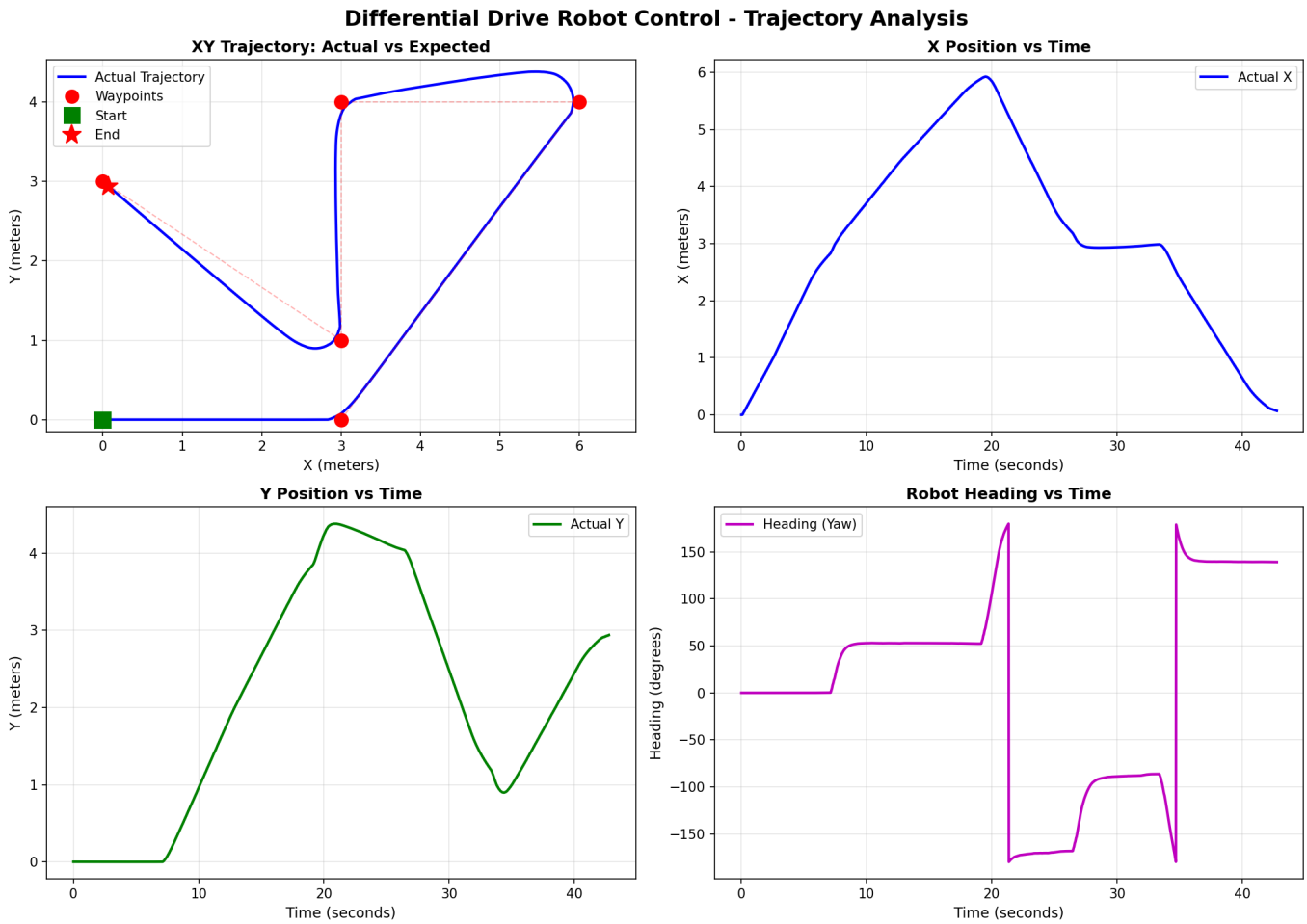
4. Observations from Data

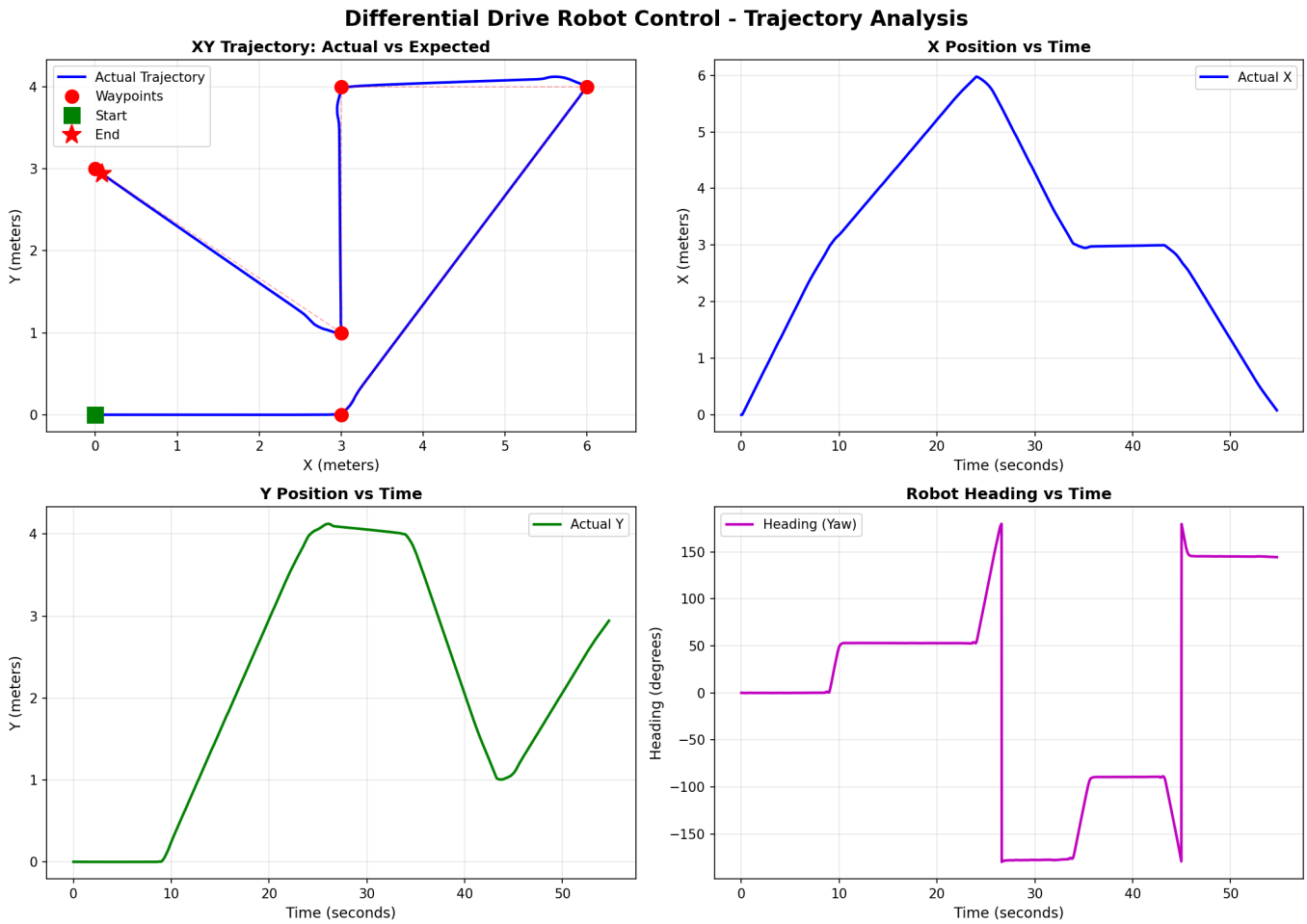
Pure Pursuit Plot



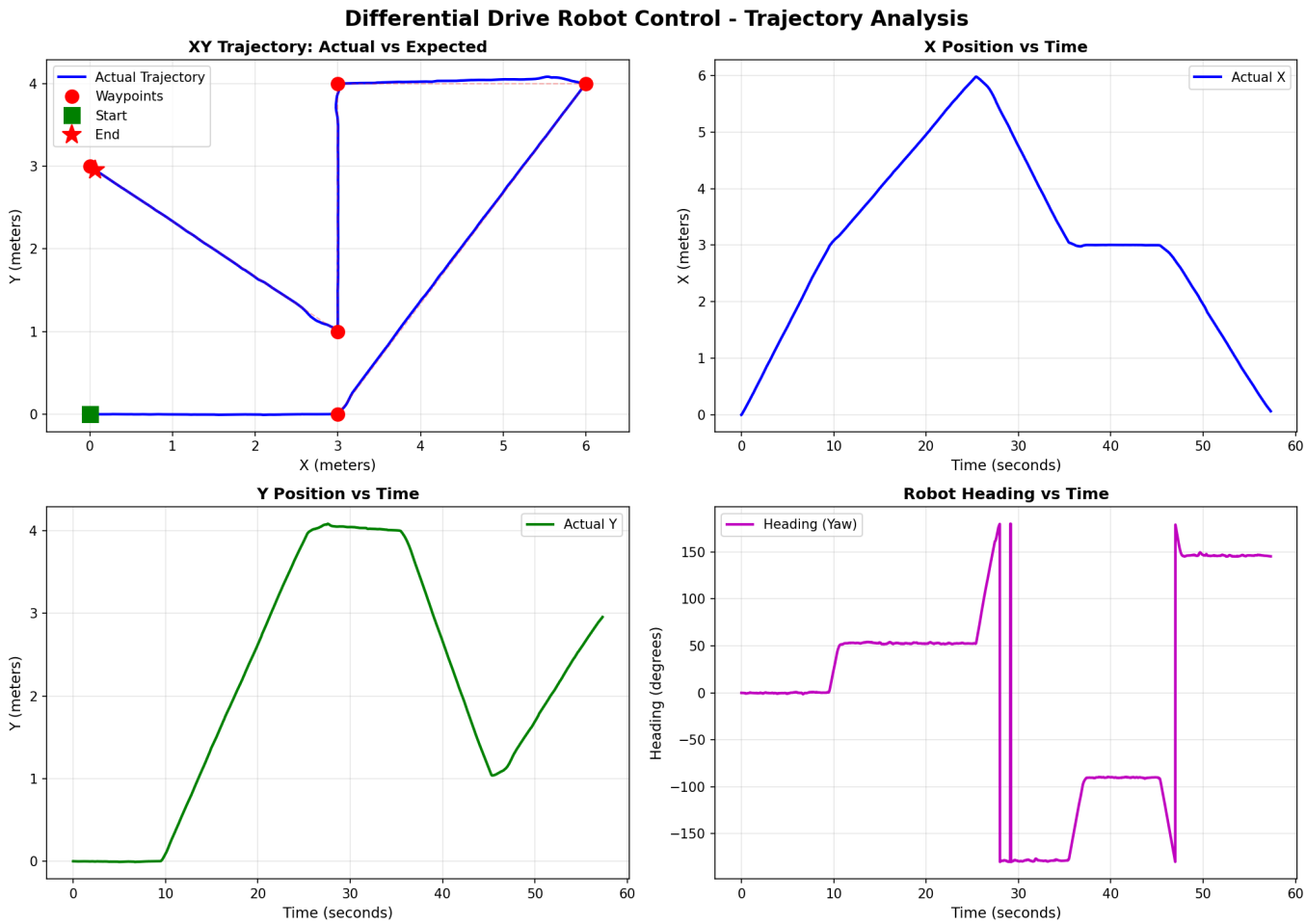
P Controller Plot



Stanley Plot


Differential Drive Robot Control - Trajectory Analysis
XY Trajectory: Actual vs Expected
Actual Trajectory
Waypoints
Start
End
Y (meters)
X (meters)
X Position vs Time
Actual X
X (meters)
Time (seconds)
Y Position vs Time
Actual Y
Y (meters)
Time (seconds)
Robot Heading vs Time
Heading (Yaw)
Heading (degrees)
Time (seconds)

MPC Plot



Behaviour Comparison

- **Pure Pursuit:** smooth and predictable on most segments with slight corner-cutting on sharp transitions.
- **P Controller:** reaches waypoints reliably but shows larger overshoot/oscillation if heading error is high.
- **Stanley:** strongest path adherence
- **MPC:** best potential smoothness after tuning, but also the most sensitive to parameter choices.

General Findings

- P and Pure Pursuit are easier to tune and debug.
- Stanley gives solid geometric path tracking.
- MPC can outperform others only when horizon and cost terms are tuned carefully.

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5. Appendix

Robot Specifications

- **Type**: Differential Drive Robot
- **Wheelbase**: [from URDF]
- **Max Linear Speed**: 0.3 m/s
- **Max Angular Speed**: 1.0 rad/s

Implementation Files

- `my_robot_controllers/pure_pursuit_controller.py`
- `my_robot_controllers/p_controller.py`
- `my_robot_controllers/stanley_controller.py`
- `my_robot_controllers/mpc_controller.py`
- Visualization script: `plot_trajectory.py`

How to Reproduce Results

1. Build the package:

```
cd ~/Homework_1
colcon build --packages-select my_robot_bringup
source install/setup.bash
```

2. Launch robot:

```
ros2 launch my_robot_bringup my_robot_gazebo.launch.xml
```

3. Run controller (in new terminal):

```
ros2 run my_robot_bringup waypoint_controller
```

4. Generate plots (after robot stops):

```
python3 ~/Homework_1/my_robot_bringup/my_robot_bringup/plot_trajectory.py
```